

2.1.1. Roots of a Quadratic Equation

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Write a program to find the roots of a quadratic equation, given its coefficients a , b , and c . Use the quadratic formula: $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

The discriminant $D = b^2 - 4ac$ determines the nature of the roots:

- If $D > 0$: Roots are real and different
- If $D = 0$: Roots are real and the same
- If $D < 0$: Roots are imaginary

Input Format:

- Three space-separated integers representing the coefficients a , b , and c , respectively.

Output Format:

- If roots are real and different, print:

```
root1 = <Root1>
root2 = <Root2>
```

- If roots are the same, print:

```
root1 = root2 = <Root1>
```

Sample Test Cases



quadratic...



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```
1 import math
2
3 a, b, c = map(int, input().split())
4
5 D = b**2 - 4*a*c
6
7 if D > 0:
8     root1 = (-b + math.sqrt(D)) / (2*a)
9     root2 = (-b - math.sqrt(D)) / (2*a)
10    print(f"root1 = {root1:.2f}")
11    print(f"root2 = {root2:.2f}")
12 elif D == 0:
13     root = -b / (2*a)
14     print(f"root1 = root2 = {root:.2f}")
15 else:
16     realPart = -b / (2*a)
17     imaginaryPart = math.sqrt(-D) / (2*a)
18     print(f"root1 = {realPart:.2f}+{imaginaryPart:.2f}i")
19     print(f"root2 = {realPart:.2f}-{imaginaryPart:.2f}i")
20
21
22
23
```

Terminal

Test cases

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Algorithm To Find Roots of a Quadratic Equation

Given a quadratic equation:

$$ax^2 + bx + c = 0$$

Step 1: Start

Step 2: Declare variables a, b, c, d, x1, x2

Step 3: Read the values of a, b, and c

Step 4: Calculate the discriminant

$$d = b * b - 4 * a * c$$

Step 5:

- If $d > 0$, calculate
 - $x1 = (-b + \text{sqrt}(d)) / (2 * a)$
 - $x2 = (-b - \text{sqrt}(d)) / (2 * a)$
 - Display Real and different roots
- Else if $d == 0$, calculate
 - $x = -b / (2 * a)$
 - Display Real and same roots
- Else
 - Display Imaginary roots

Step 6: Stop

